

1. Evaluate $\int \int_R (x+y)^2 dx dy$ where R is a parallelogram in the x, y -plane with vertices $(1,0), (3,1), (2,2), (0,1)$ using the transformation $u = x + y$ and $v = x - 2y$.
2. Evaluate $\int \int_R (x-y) \sin(x+y) dx dy$ over the tilted square region R with corners at $(0,0), (\pi,0), (\frac{\pi}{2}, -\frac{\pi}{2}), (\frac{\pi}{2}, \frac{\pi}{2})$ by doing a change of variables $u = x + y$ and $v = x - y$.
3. Find the area bounded by the parabola $y^2 = 4 - x$ and $y^2 = 4 - 4x$ as a double integral.
4. Find the area enclosed by the curves $y = 4x - x^2$ and $x = y$.
5. Find the volume of the region that lies under the paraboloid $z = x^2 + y^2$ and above the triangle enclosed by the lines $x = y, x = 0$ and $x + y = 2$ in xy -plane.
6. An image has pixel intensity given by the continuous function $f(x, y) = x^2 + y^2$ over the square region $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$. Compute the total brightness $B = \int \int_D (x^2 + y^2) dA$.
7. A two-dimensional signal has power density $P(x, y) = e^{-(x^2 + y^2)}$. The total energy of the signal over a region D is defined as $E = \int \int_R P(x, y) dA$. Evaluate the total energy E when the region D is the unit disk $D = \{(x, y) | x^2 + y^2 \leq 1\}$.
8. The upper hemisphere of radius 1 is described by $z = \sqrt{1 - x^2 - y^2}$ for (x, y) lying in the unit disk $D: x^2 + y^2 \leq 1$. Compute the surface area:

$$A = \int \int_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dA.$$

9. A robot's efficiency is given by $E(x, y) = 1 + xy$ over the rectangular region $R = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 3\}$. Compute the total efficiency:

$$T = \int \int_R (1 + xy) dA.$$

10. A rectangular CPU chip extends over the region $0 \leq x \leq 2$ cm and $0 \leq y \leq 3$ cm. The temperature distribution ($T(x, y)$ in $^{\circ}\text{C}$.) across the chip is modeled by

$$T(x, y) = 50 + 10x + 5y$$

- a. Compute the total heat measure over the chip, defined as

$$Q = \int \int_R T(x, y) dA, \text{ where } R = \{(x, y) | 0 \leq x \leq 2, 0 \leq y \leq 3\}.$$

- b. Using this result, compute the average temperature T_{avg} over the chip.

Answers:

1. 21

2. $\frac{\pi^2}{2}$

3. 8

4. 4.5

5. $\frac{4}{3}$

6. $B = \frac{2}{3}$

7. $E = \pi(1 - e^{-1})$

8. $A = 2\pi$

9. $T = 15$

10. a. $Q = 405^\circ C.Cm^2$ b. $67.5^\circ C$